***SPARKLES**: FOUNDING PRODUCT ENGINEER*

END-TO-END AI PRODUCT ENGINEERING

DECK BY PRANAV S

SKILL	WHAT I BRING	VALUE TO SPARKLES	HOW I DELIVER IT
SANDBOXED INFRA & API ARCHITECTURE	FastAPI + PostgreSQL +MongoDB systems; REST & WebSocket APIs; Redis caching (Temporary Cluster); Docker + AWS ECS/RDS/S3 deployment via Terraform. Experience with safety-sensitive, sandboxed systems.	Own **Sparkles' sandbox layer** end-to-end — isolated environments per repo, dev-validated before production push. Serve as the data & AI technical lead for this ambitious startup, with the right guardrails (code review gates, rollback triggers) built in from day one.	• At **Space Ace Inc.** (Feb 2026–present), a streaming-based startup, designing REST APIs, PostgreSQL schemas, MongoDB collections, and OAuth-based authentication using FastAPI and Python. • Containerised 5+ backend services on AWS using Terraform; dev environment validated before pushing to production. Used TypeScript and React for frontend UI. • Brings a data-first engineering mindset to every architectural decision.
LLM & CONVERSATIONAL AI INTEGRATION	RAG pipelines (ChromaDB / Pinecone), prompt engineering, few-shot frameworks, LangChain orchestration, instruction-tuned and preference-aligned LLMs, reinforcement learning, vector embeddings; worked with GPT-4o & Claude 3.5 Sonnet. Heavy AI-tool usage: Cursor, Claude Code, Codex, as part of real workflow.	Turn natural-language intent from product, design & ops teams into reviewable, **GitHub-native pull requests** — no engineering bottleneck. Non-technical input → production-ready diffs, validated safely.	• Built a **conversational chatbot** for a Statistics course textbook using RAG, vector DB, chunking, tokenization, and GPT-4o — same pattern as grounding Sparkles agents in repo context. • Applied encoders, LLM-as-judge evaluation, and few-shot training in a **News Headline Classification project**. • Academic coursework: 'AI and LLM in Data Science'. Hands-on with LangChain RAG, and vector embeddings beyond tutorial stage.
GITHUB-NATIVE INTEGRATIONS	GitHub connector & webhook design; event-driven triggers; Azure Data Factory Ingestion (5 sources); REST API integration & schema normalization across teams. Service bindings & multi-source data pipelines.	Build the GitHub-native workflow Sparkles is founded on: repo webhooks, branch lifecycle events, Pull Request triggers, reviewable diffs, and normalized data pipelines.	• At **Kentuckiana Curb Company** as BI & Strategy Analyst Intern: built Azure Data Factory ingestion pipelines from 5 sources; deployed Azure AI Foundry chatbot within a private VNet; delivered self-service BI using Power BI and Dotnet for customer ease; Postman for API testing and endpoint validation. • **Microsoft Certified Azure AI Engineer Associate (AI-102)**. • Experience building devtools-adjacent integrations: schema normalization, automated webhook-driven workflows, pull request event triggers, and API connectors at production scale across CRM & ERP systems.

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REALTIME & COLLABORATIVE SYSTEMS	Streaming & WebSocket pipelines; real-time inference focus; low-latency, high-throughput system design. Familiarity with Twilio/LiveKit-style streaming and Cloudflare Workers-compatible edge architectures like CDN.	Ship **Sparkles' live preview and collaborative product surfaces** — sub-second code change previews, WebSocket-powered branch sync, realtime diff rendering.	• Built real-time computer vision system (30 fps on Raspberry Pi CM4) at **Spark Minda**, a telematics company in Bangalore — same low-latency, high-throughput constraints as voice pipelines. • Applied BLEU score evaluation for speech-to-text output quality assessment. • Comfortable with Twilio/LiveKit-style streaming architectures.
ML PIPELINE & DATA SCIENCE	XGBoost & scikit-learn for classification/regression; ARIMA, Prophet & LSTM (PyTorch) for time-series demand forecasting; DistilBERT for NLP sentiment; Isolation Forest for anomaly detection; ETL pipelines with Apache Airflow; feature engineering & model evaluation.	Build the safety layer for existing codebases: change-risk scoring per PR, anomaly detection on outbound diffs, and **turning every merged commit into a training signal**.	• Authored **full ML pipeline mapped to Sparkles**: XGBoost change-risk classifier trained on PR history, diff size & file-touch frequency; Logistic Regression PR approval scoring using edit type & review patterns; Isolation Forest anomaly detection to flag out-of-pattern diffs before production. • DistilBERT NLP for intent classification: parsing non-technical requests into structured code instructions; diff summarization to auto-generate plain-English PR descriptions; commit message parsing to feed structured signals into the risk model. • ETL pipelines to extract, clean & warehouse structured data from commits, PRs & agent interactions, Sparkles' training warehouse. • Grading Assistant — 'ML Principles' & 'Applied Statistical Learning', Rutgers University.
PRODUCT OWNERSHIP	Authored Sparkles backend/AI architecture deck; maps current-state gaps to concrete build opportunities; productionized end-to-end projects; **builds the stand, not just works in it**.	Thinks like a founder, not just an engineer, identifies what to build and why, then builds it. Driven to turn raw data and AI into automated systems, not just tools. Brings the "build the stand, don't just work at it" mentality.	• Passionate about automating the full ML lifecycle using agentic AI and structured protocols — building custom solutions, not just integrating existing ones. • Moves fast without lowering the bar: productionizes end-to-end, not just integrates existing tools and authored Sparkles architecture deck. • **TEDxGCT Curator**, took ownership of curating new ideas worth spreading.